Reliable ABC model choice via random forests

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Outline

1. Problem Formulation
2. Method
3. Data
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ABC for Model Choice: Principle

Setting. We consider a collection of competing models

$$m \in \{1, \ldots, M\},$$

with prior probabilities $\pi(m)$ and model-specific priors $\pi(\theta \mid m)$. The likelihood $f(x \mid m, \theta)$ is assumed to be *intractable*, but simulation from the generative model is possible.

Bayesian objective. For the observed dataset x^0, the goal is to infer the posterior model probabilities

$$\pi(m \mid x^0).$$

ABC principle. The likelihood $f(x^0 \mid m, \theta)$ is approximated through the proximity between x^0 and pseudo-data $x(\theta)$ simulated from $f(x \mid m, \theta)$. In practice, the comparison is performed using:

- a vector of summary statistics $S(x)$,
- a distance $d(\cdot, \cdot)$,
- a tolerance $\varepsilon > 0$.

ABC Model Choice: Algorithm and Limitations

Reference table.

$$(m, \theta, x) \sim \pi(m)\,\pi(\theta \mid m)\,f(x \mid m, \theta), \qquad \text{stored as } (m, S(x)).$$

ABC approximation. Posterior probabilities are estimated by k-NN:

$$\widehat{\pi}(m \mid x^0) = \frac{1}{k} \sum_{i \in \mathcal{N}_k(x^0)} \mathbf{1}\{m_i = m\}, \quad \mathcal{N}_k(x^0) : k \text{ nearest to } S(x^0).$$

Decision rule.

$$\hat{m} = \arg\max_m \widehat{\pi}(m \mid x^0).$$

Limitations.

- Strong dependence on $S(x)$
- Curse of dimensionality (k-NN)
- Unreliable posterior probabilities

Consequence. These issues motivate a reformulation of ABC model choice as a *classification problem*.

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ABC Model Choice as a Classification Problem

Limitation of standard ABC. Classical ABC model choice approximates posterior probabilities $\pi(m \mid S(x^0))$ using local methods, which are unreliable in high dimension and strongly depend on the choice of summary statistics.

Reformulation. ABC model choice is recast as a supervised classification problem:

$$m \longleftarrow S(x),$$

where the model index m is predicted from the summary statistics $S(x)$.

Random Forest classifier. The mapping $S(x) \mapsto m$ is learned using a Random Forest classifier, trained on the ABC reference table.

Training data. The classifier is trained on simulated pairs

$$(m, S(x)) \sim \pi(m)\,\pi(\theta \mid m)\,f(x \mid m, \theta),$$

generated independently of the observed data x^0.

Objective. The aim is to approximate the (infeasible) Bayesian classifier

$$m^\star(x) = \arg\max_m \pi(m \mid x),$$

which minimizes the integrated 0–1 loss (prior error rate).

ABC-RF: Posterior Probability via a Second Random Forest

Model choice (RF classification). A Random Forest classifier trained on the reference table returns the selected model

$$\hat{m}(S(x^0)).$$

The classification vote is *not* a posterior probability.

Key identity. The posterior probability that the selected model is the true one satisfies

$$\mathbb{P}\Big(m = \hat{m}(S(x^0)) \mid S(x^0)\Big) = 1 - \mathbb{P}\Big(\hat{m}(S(Y)) \neq m \mid S(Y) = S(x^0)\Big). \tag{1}$$

Second RF (regression). Define the misclassification indicator

$$Z = \mathbf{1}\{\hat{m}(S(Y)) \neq m\}.$$

A second Random Forest, used in *regression* mode, learns

$$q(s) \approx \mathbb{E}[Z \mid S(Y) = s] = \mathbb{P}(\hat{m}(S(Y)) \neq m \mid S(Y) = s),$$

using (out-of-bag) predictions on the reference table.

Posterior probability estimate.

$$\widehat{\mathbb{P}}\Big(m = \hat{m}(S(x^0)) \mid S(x^0)\Big) = 1 - q(S(x^0)).$$

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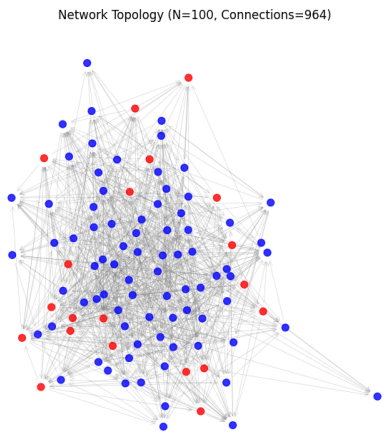
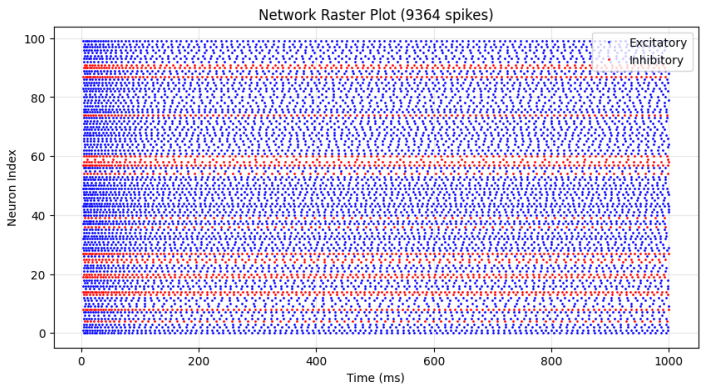
Data: Simulated Neural Activity

Inference Goal: We observe the **raster plots** ($X \in \{0,1\}^{N \times T}$, simulated by *Brian2* package) and **aim to recover the hidden connectivity model** ($m \in \{0,1,2,3\}$, generated by *NetworkX* package).

Model 1: Erdos-Renyi

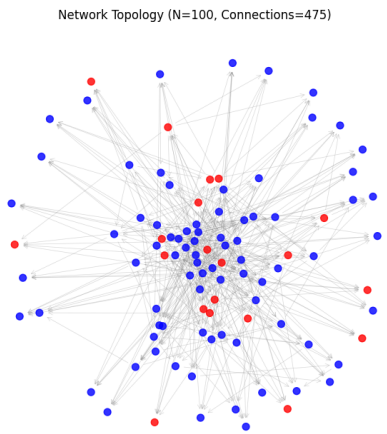
(Random Binomial Connectivity)

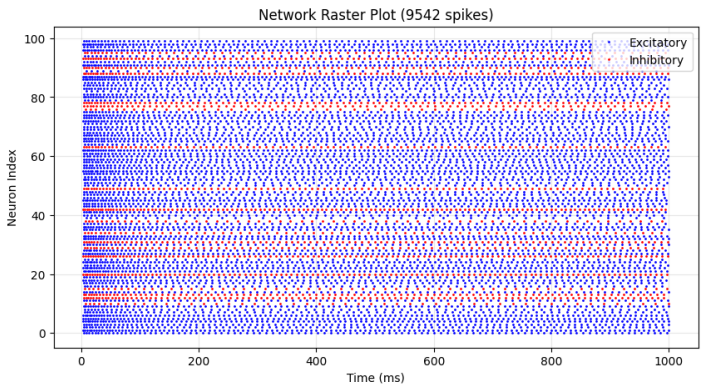




Model 2: Barabasi-Albert

(Hub Structure / Scale-Free)





Summary Statistics

Feature extraction. Spike trains are mapped to a low-dimensional representation using summary statistics computed from the raster plot.

Population-level statistics

- Mean firing rate (Hz)
- Population Fano factor
- Mean pairwise correlation

Single-neuron statistics

- Coefficient of variation (CV)
- Local variation (LV)

Total: 5 ePhys 1D-features + 3 LDA
$\Rightarrow$ 8 scalar features

Implementation

computed using the `Elephant` package.

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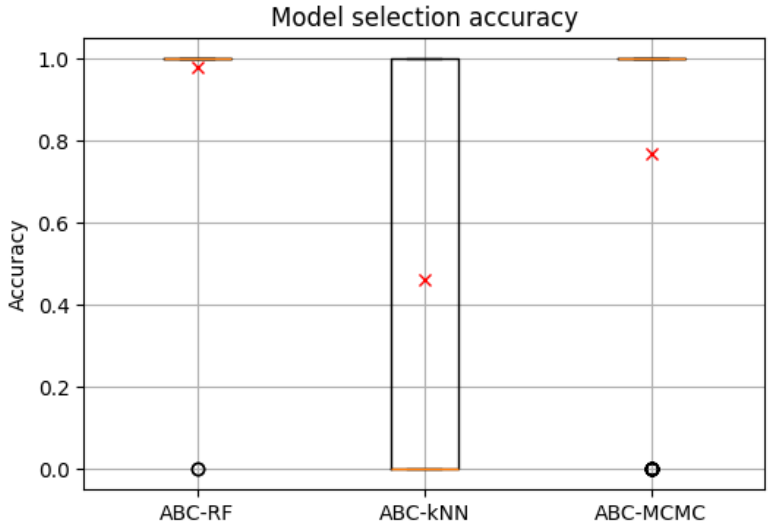
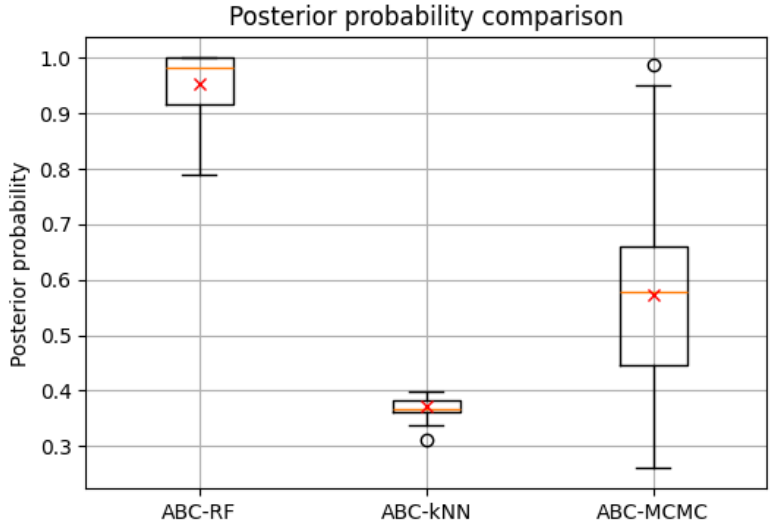
Benchmark Results: ABC-RF vs. Competitors

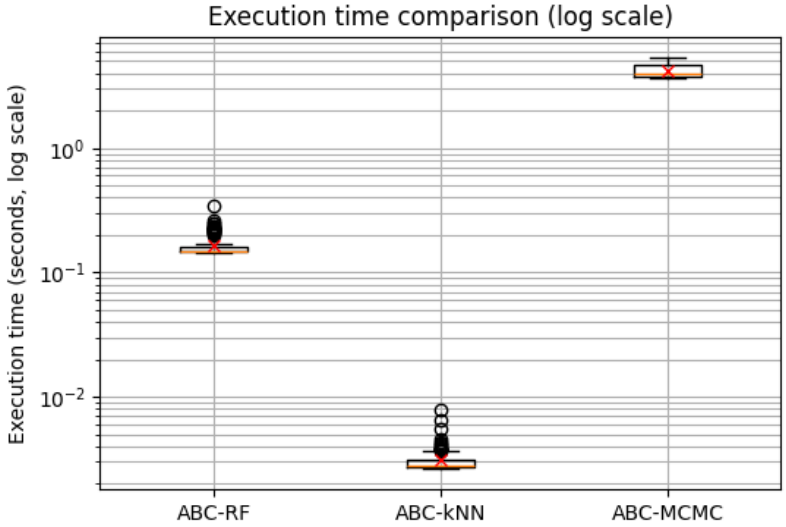
1. Model Selection Accuracy



2. Posterior Confidence



3. Execution Time



Key Findings:

- **Accuracy:** ABC-RF ($\approx 98\%$, red cross) significantly outperformed competitors.
- **Confidence:** ABC-RF provides sharper posterior estimates (not really relevant since accuracies are comparable).
- **Efficiency:** ABC-RF is orders of magnitude faster than MCMC, offering the best **quality-to-cost trade-off**.

Thank you for your attention!

Questions?

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Discussion: Strengths & Limitations

Theoretical Advantages (+)

- **Computational Efficiency:** Unlike k-NN, ABC-RF does not rely on local density estimation or a tolerance ε. It avoids the *curse of dimensionality* and often requires fewer simulations.
- **Robustness to Summary Stats:** RF is naturally robust to noisy or redundant features in $S(x)$.
- **Synergy with LDA:** We can combine raw $S(x)$ with LDA axes. RF captures non-linear interactions from $S(x)$ while benefiting from the optimal class separation of LDA.
- **Shallow vs. Deep Learning:** RF is less data-hungry than Deep Learning (e.g., MLPs) and harder to overfit, making it ideal for expensive simulation contexts.

Limitations & Open Issues (-)

- **"Winner Takes All" Problem:** Since this is a classification task, the RF *always* predicts a winner, even if all models are poor explanations for the data.
- **Goodness-of-Fit Check:** Reliability requires a 3-step process: (1) Select Model ($\hat{m}$) (2) Estimate Posterior Probability (3) Visual Check: Project observed data $S(x^0)$ onto LDA axes to ensure it falls within the training distribution.
- **Dependence on $S(x)$:** While robust to noise, the method cannot invent missing information. If $S(x)$ is non-informative, the classifier will fail.

ABC-RF: Reference Table and Model Choice

Algorithm 1: ABC reference table

For $i = 1, \ldots, N_{\text{ref}}$:

$$m \sim \pi(m), \qquad \theta \sim \pi(\theta \mid m), \qquad x \sim f(x \mid m, \theta),$$

store $(m, S(x))$.

Algorithm 2: Model choice (RF classification)

Train a Random Forest classifier to predict m from $S(x)$. For the observed data, predict

$$\hat{m} = \hat{m}(S(x^0))$$

by majority vote over the trees.

ABC-RF: Posterior Probability Estimation

Algorithm 3: Posterior probability estimation

The classification vote does not provide posterior probabilities. Using out-of-bag predictions, define the misclassification indicator

$$Z = \mathbf{1}\{\hat{m}(S(x)) \neq m\}.$$

Train a Random Forest regression to estimate

$$q(s) = \mathbb{P}(Z = 1 \mid S(x) = s) = \mathbb{P}(\hat{m}(S(x)) \neq m \mid S(x) = s)\,.$$

For the observed data, return

$$\widehat{\mathbb{P}}\Big(m = \hat{m}(S(x^0)) \mid S(x^0)\Big) = 1 - q(S(x^0)).$$